Trial SET #6 (math) :



- Don't use calculator and any other materials, including formulae and tables;
- You have only 60 minutes to complete the test;
- Try to do your best and Good Luck!!!

You should be familiar with:

- word problems;
- angles in triangles;
- number properties;
- combinatorics;
- integrals;
- trigonometry;
- arithmetic sequences;
- roots;
- areas in quadrilaterals;
- circles;
- logarithms;
- derivatives;

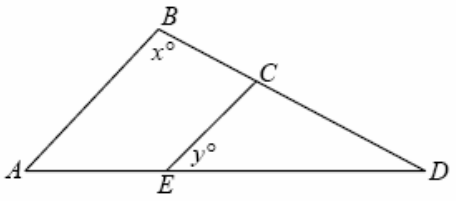
Name of the student _____

1. Kim purchased n items from a catalog for \$8 each. Postage and handling charges consisted of \$3 for the first item and \$1 for each additional item. Which of the following gives the total dollar amount for Kim's purchase, including postage and handling, in terms of n ?

(A) $8n + 2$ (B) $8n + 4$ (C) $9n + 2$ (D) $9n + 3$ (E) $9n + 4$

2. If water is leaking from a certain tank at a constant rate of 1,200 milliliters per hour, how many seconds does it take for 1 milliliter of water to leak from the tank?

A. 1/3 B. 1/2 C. 2 D. 3 E. 20



Note: Figure not drawn to scale.

3. In the figure above, if AB || CE, $CE = DE$, and $y = 45$, then $x =$

(A) 45 (B) 60 (C) 67.5 (D) 112.5 (E) 135

4. The farm has chickens that lay only white eggs and brown eggs. On a certain day, the chickens lay a total of 200 eggs in which the ratio of white eggs to brown eggs is 7:3. If the ratio of white eggs to brown eggs is to be changed to 2:1 by adding only brown eggs, how many brown eggs must be added?

A. 10; B. 15; C. 18; D. 24; E. 30;

5. In the fifth grade at Parkway elementary school there are 420 students. 312 students are boys and 250 students are playing soccer. 86% of the students that play soccer are obviously boys. How many girl student are in Parkway that are not playing soccer?

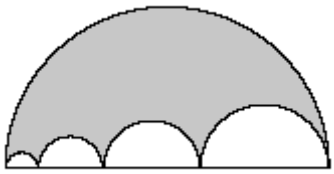
A. 69. B. 73. C. 81. D. 91. E. 108.

6. A student responded to all of the 22 questions on a test and received a score of 63.5. If the scores were derived by adding 3.5 points for each correct answer and deducting 1 point for each incorrect answer, how many questions did the student answer incorrectly?

(A) 3
(B) 4
(C) 15
(D) 18
(E) 20

7. $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + 7^2 - 8^2 + \ldots - 1000^2 = ?$

A. -500500 B. 25050 C. -500 D. -2500 E. -50500

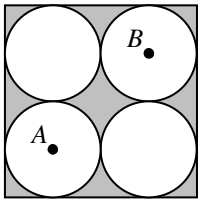


8. The figure above consists of four small semicircles in a large semicircle. If the small semicircles have radii of 1, 2, 3, and 4, what is the perimeter of the shaded region?

(A) 10π
(B) 20π
(C) 40π
(D) 60π
(E) 100π

9. Karen is ordering a hamburger. The KFS offers 4 different kinds of bred, 5 different condiments, 3 different kinds of cheese and 2 types of meat. If Karen selects one type of bread, one condiment, and she has to choose between cheese and meat, in how many ways can she order her burger?

A. 100 B. 120 C. 14 D. 180 E. 15



10. As the figure above shows, four identical circles are inscribed in a square. If the distance between centers of circle *A* and *B* is $4\sqrt{2}$, what is the area of the shaded region?

(A) $256 - 72\pi$ (B) $256 - 64\pi$ (C) $256 - 32\pi$
(D) $128 - 32\pi$ (E) $64 - 16\pi$

11. A is even and B is odd. Which of the following expressions can't be an integer?

A. $(A+1)/(B+1)$ B. $(A+1)/B$ C. $(B-1)/(A^2)$

D. $(A)/(64)$ E. $(A^4)/(AB)$

12. $f(x) = 2x + 3$, $g(x) = x^2$. Find $f[g(x)]$.

A. $(2x+3)^2$ B. $2x^2+3$ C. $4x^2+3$ D. $2x+9$ E. 25

13. If $8^{2x} = 4^{-1}$ then what is the value of 8^{-x}.

A) $2\sqrt{2}$ B) 2 C) $\sqrt{2}$ D) $\frac{1}{4}$ E) $\frac{1}{2}$

14. If $8^{a-b} = 32^{a+b}$ then find the value of $\frac{2b-a}{7b+a}$.

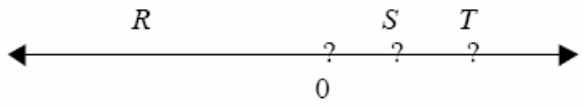
A) 2 B) 4 C) 6 D) 8 E) 16

15. Of the total amount that Jill spent on a shopping trip, excluding taxes, she spent 50 percent on clothing, 20 percent on food, and 30 percent on other items. If Jill paid a 4 percent tax on the clothing, no tax on the food, and an 8 percent tax on all other items, then the total tax that she paid was what percent of the total amount that she spent, excluding taxes?

A. 2.8% B. 3.6% C. 4.4% D. 5.2% E. 6.0%

16. For any numbers a and b, $a \blacklozenge b = a^2 + b - 2ab$. If $a \blacklozenge b = 0$, which of the following CANNOT be a value of b?

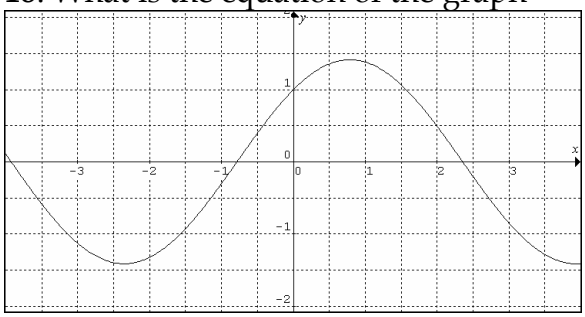
A. 2 B. 3 C. -1/2 D. 1/2 E. -3/2



17.
The number line shown contains three points *R*, *S*, and *T*, whose coordinates have absolute values *r*, *s*, and *t*, respectively. Which of the following equals the average (arithmetic mean) of the coordinates of the points *R*, *S*, and *T* ?

A. s B. $s + t - r$ C. $(r-s-t)/3$ D. $(r+s+t)/3$ E. $(s+t-r)/3$

18. What is the equation of the graph



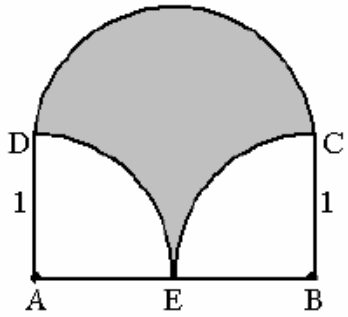
A. sinx + cosx B. sinx – cosx C. sin2x + cos2x
D. sin2x - cos2x E. 2sinx + 2cos x

19. If $0 < a < 1 < b < c$, what are the signs of $\log_b a$, $\log_b c$, $\log_b b$ and $\log_c b$ respectively?

A) - - + + B) - - + - C) - + - + D) + + + + E) - + + +

20. Let $P(x) = 3x^4 - (m-1)x^3 + 2(n-1)x + 6$ and $Q(x) = ax^4 - bx^2 + 6x + c$.
What is m + n, if P(x) = Q(x)?

A. 3 B. 4 C. 5 D. 6 E. 7



21. ABCD is a rectangle and given semicircle has a diameter *DC*. If two quarter circle centered at A and B have radii 1 cm then find the shaded area.

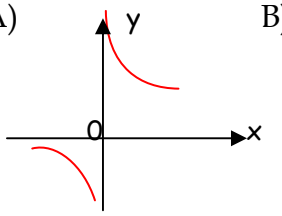
A) $\pi+1$ B) 2 C) $3-\pi$ D) π E) 4

22. If x is a number satisfying $2 < x < 3$ and y is a number satisfying $7 < y < 8$, which of the following expressions will have the largest value?

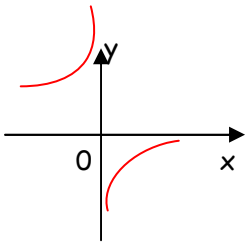
(A) x^2y (B) xy^2 (C) $5xy$ (D) $\frac{4x^2y}{3}$ (E) $\frac{x^2}{y}$

23. Which one of the following graphs represent log x – log y = 1.

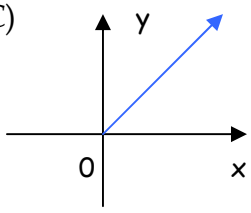
A)



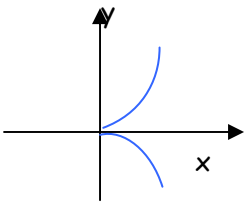
B)



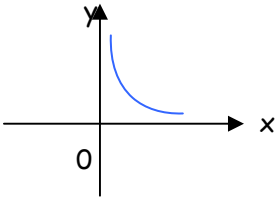
C)



D)



E)



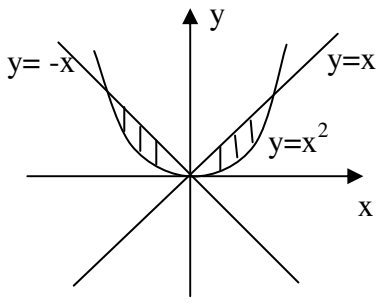
24. If a 3-digit integer is selected at random from the integers 100 through 199, inclusive, what is the probability that the first digit and the last digit of the integer are each equal to one more than the middle digit?

A. 2/225 B. 1/111 C. 1/110 D. 1/100 E. 1/50

25. The first three terms of a sequence are 2, 7, and 22. After the first term, each consecutive term can be obtained by multiplying the previous term by 3 and then adding 1. If the sequence continues to be expanded, what will be the sum of the tens digit and the units digit of the thirty-fifth term in the sequence?

a. 2
b. 4
c. 7
d. 9
e. 13

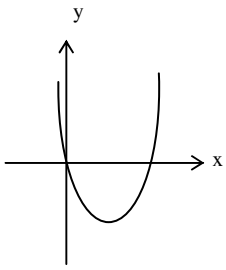
26. In the figure below, find the area of the shaded region.



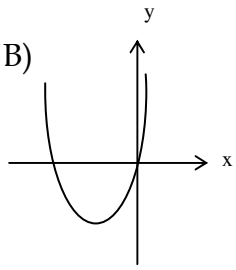
A) $\frac{1}{3}$ B) $\frac{1}{6}$ C) $\frac{4}{3}$ D) $\frac{5}{3}$ E) $\frac{5}{6}$

27. Which one of the followings may be the graph of the function $y = x^4 - 2x^2$?

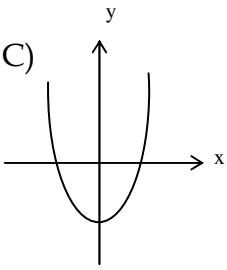
A)



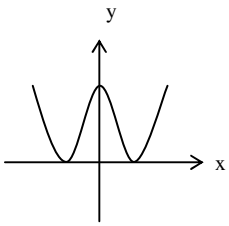
B)



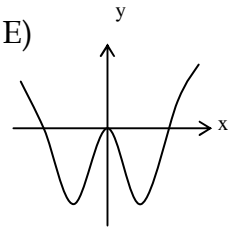
C)

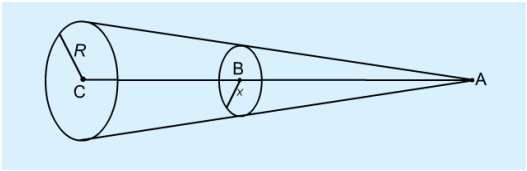


D)



E)





28. The volume of a cone is *S* **H*/3, where *S* is the area of the base and *H* is the height. If *AB* = *BC*, what is the ratio of the volume of the top half of the cone (with *AB* as height) to the volume of the bottom half of the cone (with *BC* as height)?

a. 1 : 3
b. 1 : 4
c. 1 : 6
d. 1 : 7
e. 1 : 8

29. $\frac{cosx}{1-sinx} + \frac{cosx}{1+sinx} = ?$

A) $2\csc x$ B) 2 C) cosx D) $2\sec x$ E) sinx

30.

*	a	b	c	d
a	c	d	a	b
b	d	c	b	a
c	a	b	c	d
d	b	a	d	c

The table above defines operation * on the set { a , b , c , d }. According to the definition of the operation, find the inverse of d.

A) a B) b C) c D) d E) it has no inverse